
The Smart Flood Evacuation Route Planning System

System Plan

Revision 4

Revision History

Revision 1

2026/04/20

- Base Road Network Map
- Shortest Path Algorithm
- Custom Road Network

Revision 2

2026/05/19

- Manual Pinning / Drop a Pin
- Address-Based Pinning / Geocoding
- Current Location Pinning

Revision 3

2026/05/22

- API Integration

Revision 4

2026/05/26

- Flood Susceptibility Zones

System Specification

Requirement Definition

The Smart Flood Evacuation Route Planning System is a web-based GIS platform designed to support disaster prevention and emergency evacuation decision making. The system integrates flood potential maps, real-time rainfall information, road network analysis, hospitals, and evacuation

shelters. Users can simulate different rainfall scenarios and obtain optimal evacuation routes while avoiding flooded areas.

Development Environment

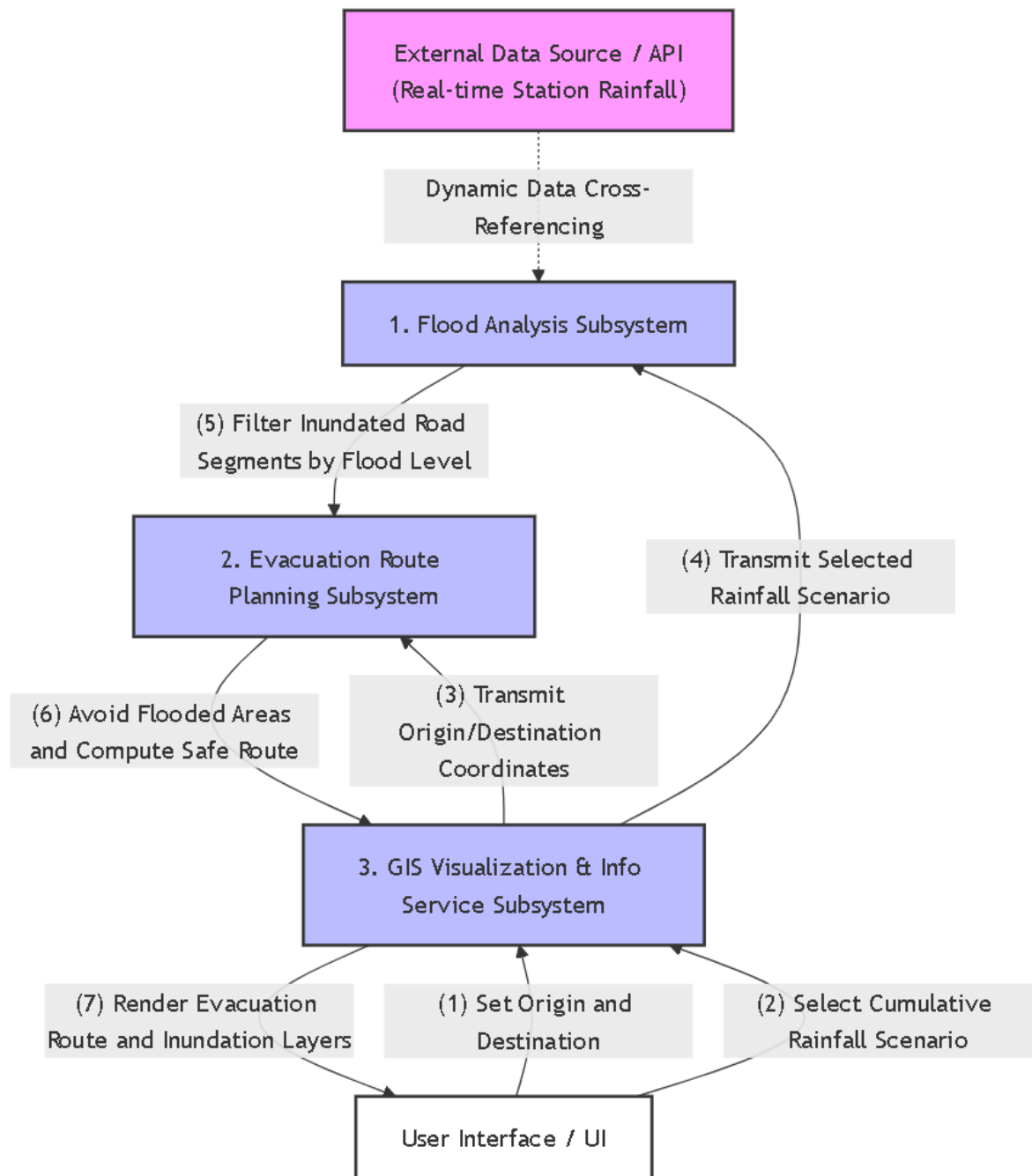
- Target web browsers:
 - Google Chrome
 - Microsoft Edge
- Development tools
 - Backend Framework: Python Flask
 - Frontend Core: HTML5, CSS3, JavaScript (ES6+)
 - Web Mapping Library: Leaflet.js
 - Geospatial Data Analysis: GeoPandas
 - Network & Routing Processing: OSMnx, NetworkX
- Data Standard
 - Follow ISO/TC211 and OGC standards to process related metadata.
 - Conform to the coordinate systems in Taiwan.

System Scope

The Smart Flood Evacuation Route Planning System includes 3 subsystems as below:

1. Flood Analysis and Emergency Location Subsystem
2. User Interface and Public Information Service Subsystem
3. Dynamic Evacuation Route Planning Subsystem

The main functions and data dataflow for each subsystem are planned according to the contract specifications. The following diagram is the relationship between subsystems.

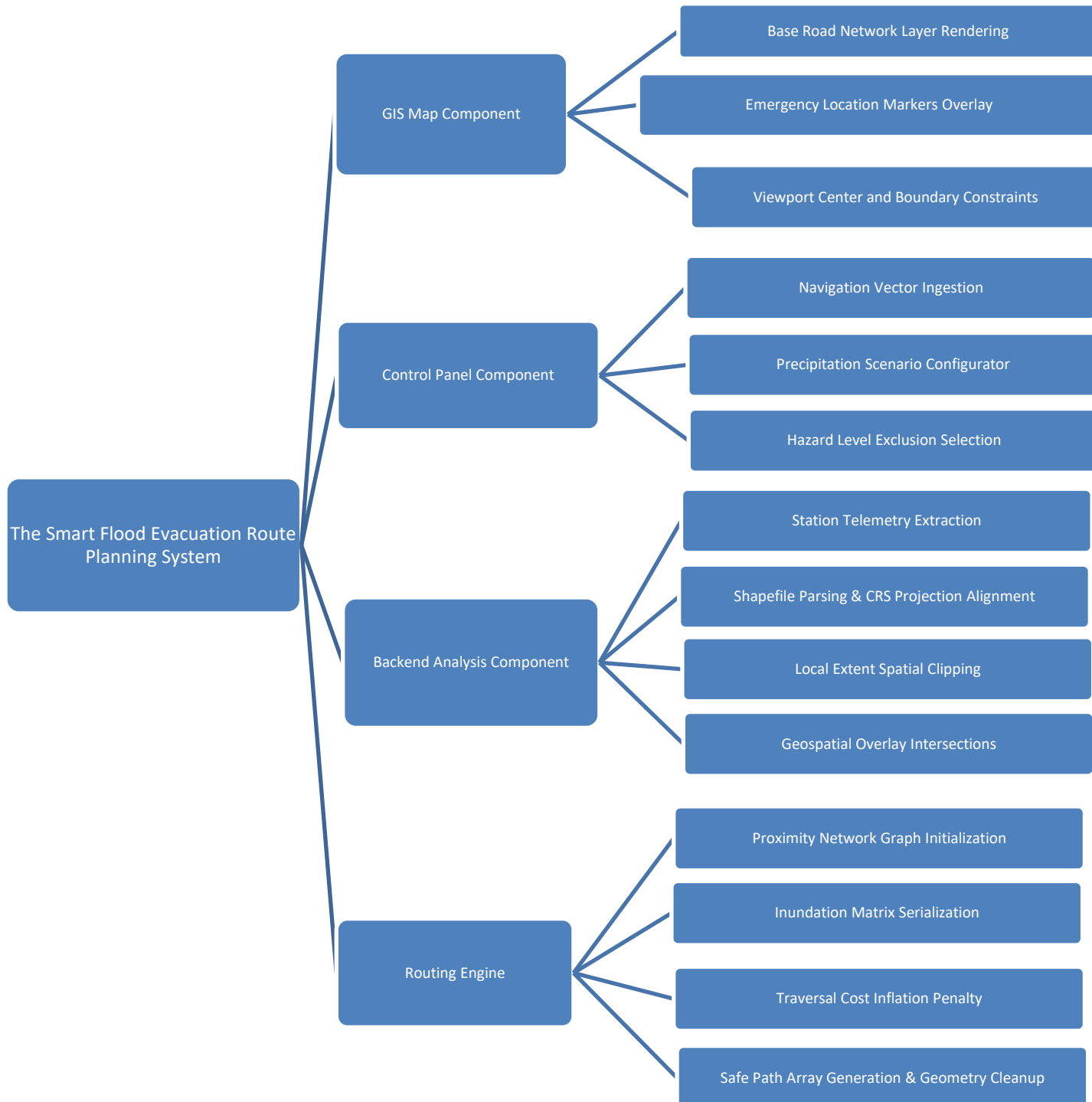


System Development Plan

The system development plan focuses on defining operational and analytical system requirements. We conduct a more detailed system structures and user interface design necessary to support work activities.

System A

System Structure



Conceptual UI Elements

The user interface is engineered as an integrated, single-page GIS cockpit to guarantee zero-barrier execution during high-stress disaster deployment. The layout is partitioned into a left-side functional control panel for scenario variables and a center map canvas for geospatial visualization.



Main Functions

The main task in this working item is to manage reporting water-level telemetry data and value-added inundation maps. The Smart Flood Evacuation Route Planning System has the following main functions:

1. Map View (M0)

The main map view is initialized globally when the web application is accessed via a desktop web browser. By default, the system utilizes a persistent two-column layout tailored for desktop displays: the control panel is permanently positioned on the left side of the screen, while the interactive Leaflet map canvas occupies the remaining right portion of the viewport to display spatial routing analysis.

UI Elements

M0-01	Activates the address search and geocoding input field on the left panel. Captures the user's destination string to resolve geographic coordinates, preparing parameters for the routing engine.
M0-02	Triggers browser-side HTML5 Geolocation. Pressing this button will center the map on the user's current GPS location and assign it as the route origin point.
M0-L	Toggles the floating overlay control menu Popup M0-L to manage geospatial map layers.
M0-P	Triggers a POST request to the backend <code>/get_route</code> endpoint, passing start, end, scenario, and avoid_levels to execute the custom weight_function path analysis.

2. Custom Overlays & Layer Management (M0-L)

A static control menu embedded within the panel for selecting the geospatial layers to be shown in the map view M0. Clicking on the respective layer items or checkboxes will instantly toggle their visibility on the map. When a specific flood scenario is selected, backend layers are fetched via `/get_flooding_zones` and rendered dynamically onto the active map canvas. Available independent layers:

Available independent layers:

Evacuation Shelters & Hospitals	Renders the 17 geo-validated emergency facilities defined in the backend EMERGENCY_LOCATIONS registry array.
Real-time Rainfall Stations	Plots markers for Xinzhuang, Banqiao, and Guishan telemetry stations, embedding dynamic rainfall depth information (rain_all_day).

Available mutually exclusive layers:

(Exactly one of these layers can be active at any time.)

Standard street map	The default lightweight OpenStreetMap (OSM) base network layer.
Inundation Scenario Layer	Renders the custom vector polygons filtered by load flooding zones, displaying color-coded flood depths (Grid Codes 1 to 6) ranging from 0.1m to over 3.0m.

3. Map Query & Route Output (M0-P / M0-R)

This layer appears after the user configures the evacuation parameters and executes the path planning algorithm. Upon calculating the dynamic matrix penalties, the web interface retrieves the query array and draws vector markers onto the Leaflet canvas.

● UI Elements / Map Markers

Origin & Destination Markers M0-P01	Pinpoints the start (Green Marker) and final end-point (ex. Shelter) (Red Marker) selected by the user on the map.
Inundated Road Blockage Marker M0-P02	Identifies segments caught in the spatial join intersection (gpd.sjoin). Contains an icon and a text annotation displaying the simulated depth text mapping (e.g., "Water level deep enough to stall vehicles").

Optimized Safe Route Polyline M0-R01	The resulting hazard-free path array (route_points) generated by nx.shortest_path, plotted as a distinct high-visibility vector line avoiding all selected avoid_levels.
--------------------------------------	--

User Authorization

Main Functions

The system initialization routine is executed the first time the web platform is launched in a browser session. During startup, the Flask backend validates connection integrity with external meteorological web services and maps regional topology dependencies.

Connection & Data Ingestion Verification (L0)

- UI Elements

L0-01	Title of the Web Application shown on the header banner.
L0-02	CWA API Authorization Token verification state check.
L0-03	Spatial Data Folder verification check.
L0-04	Error message / Status log box shown on the diagnostic panel if any dependency fails.
L0-05	Map Initialization Trigger

Possible error messages for L0-4:

L0-M01	Network connection unstable. Activating local backup database.
L0-M02	Target Inundation Shapefile missing. Please verify system data directories.
L0-M03	OSM Network Graph failed to initialize within the specified bounding radius.